

Civil Engineer to Data Scientist (with GenAI) - 8 Month Roadmap

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A Complete 8-Month Transition Roadmap with Learning Resources

This guide provides an 8-month structured plan for Civil Engineers to transition into Data Science and Generative AI roles. It focuses on practical skills, project-based learning, and curated YouTube, Coursera, Udemy, and LinkedIn Learning resources.

Month 1: Core Foundations & Data Literacy

Goals: Learn basic programming, statistics, and data handling. Topics: Math & Stats, Excel, Python basics, Pandas, NumPy. Resources: CampusX 100 Days of Python (Days 1-25), Udemy Python Bootcamp by Jose Portilla, LinkedIn Learning Analytics by Robin Hunt. Mini Project: Sales dataset dashboard in Excel.

Month 2: Python for Data Analysis & Visualization

Goals: Learn to clean, manipulate, and visualize data. Topics: Pandas, NumPy, Matplotlib, Seaborn, EDA. Resources: CampusX 100 Days of Python (Days 26-60), Kaggle Python course, LinkedIn Python for Data Analysis. Mini Project: Analyze HR dataset and create insights report.

Month 3: SQL and Power BI / Tableau

Goals: Learn data querying and BI visualization. Topics: SQL queries, Joins, Power BI/Tableau dashboards, KPIs. Resources: CampusX SQL, LinkedIn Learning Power BI, Coursera SQL for Data Science. Mini Project: Power BI dashboard for HR/Sales analysis.

Month 4: Machine Learning Foundations

Goals: Learn ML algorithms and fundamentals. Topics: Regression, Classification, Supervised/Unsupervised learning, Evaluation metrics. Resources: CampusX 100 Days ML (Days 1-40), Krish Naik ML Roadmap 2025, Coursera ML by Andrew Ng. Mini Project: Customer churn or price prediction model.

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Month 5: Advanced ML & Deployment

Goals: Model optimization, tuning, and deployment. Topics: Feature engineering, XGBoost, FastAPI, Git/GitHub. Resources: Krish Naik FastAPI, LinkedIn MLOps Fundamentals, CampusX 100 Days ML (Days 40-100). Mini Project: Deploy a classification model via FastAPI.

Month 6: Deep Learning & Neural Networks

Goals: Understand DL fundamentals and frameworks. Topics: Neural Nets, CNN, RNN, TensorFlow, Keras. Resources: CampusX 100 Days DL (Days 1-50), Coursera Deep Learning Spec, LinkedIn Neural Networks. Mini Project: Image classification using CNN.

Month 7: Generative AI & LangChain

Goals: Work with LLMs and RAG. Topics: Prompt Engineering, Embeddings, RAG, LangChain, Vector DBs. Resources: CampusX GenAI with LangChain, Krish Naik GenAI Projects, Coursera GenAI with LLMs. Mini Project: RAG chatbot over PDFs using LangChain.

Month 8: Agentic AI, LangGraph & Portfolio

Goals: Learn LangGraph & build full GenAI systems. Topics: LangGraph agents, FastAPI integration, Streamlit/Gradio UI. Resources: CampusX LangGraph Tutorials, Krish Naik FastAPI Projects, LinkedIn OpenAI API Apps. Projects: RAG chatbot, Agent workflow automation, Streamlit AI Dashboard. Portfolio Goal: GitHub, LinkedIn, Portfolio website with 34 AI projects.

Final Summary and Learning Reflection

Summary: In 8 months, you'll progress from Data Analysis to ML to Generative AI, using CampusX, Krish Naik, Coursera, and LinkedIn Learning resources. My Real Learning Journey: I personally mastered everything by following just 3 playlists from CampusX: 1. 100 Days of Python 2. 100 Days of Machine Learning 3. 100 Days of Deep Learning Then continued with Generative AI with LangChain and LangGraph + FastAPI tutorials. Final Advice: - Focus on consistency and projects. - Certificates matter less than your GitHub portfolio. - Keep learning, sharing, and

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building publicly on LinkedIn.